

星际介质天文学作业

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2024 年 3 月 15 日

21.1 由 R_V 的定义:

$$R_V = \frac{A_V}{A_B - A_V}$$

如果 $A(\lambda) \propto \nu \propto \frac{1}{\lambda}$, 则:

$$R_V = \frac{1/\lambda_V}{1/\lambda_B - 1/\lambda_V} = \frac{1/5470 \text{ \AA}}{1/4405 \text{ \AA} - 1/5470 \text{ \AA}} = 4.136$$

21.2 由 R_V 的定义:

$$R_V = \frac{A_V}{A_B - A_V}$$

如果 $A(\lambda) \propto \nu^\beta \propto \lambda^{-\beta}$, 则:

$$R_V = \frac{\lambda_V^{-\beta}}{\lambda_B^{-\beta} - \lambda_V^{-\beta}} = 3.1$$

代入 $\lambda_B = 4405 \text{ \AA}$ 、 $\lambda_V = 5470 \text{ \AA}$, 解得:

$$\beta = 1.291$$

21.3 (a) 考虑单位长度尘埃和氢原子的质量, 单位长度的质量等于粒子的质量乘粒子的柱密度:

$$\frac{M_{\text{dust}}}{M_{\text{H}}} = \frac{m_{\text{dust}} N_{\text{dust}}}{m_{\text{H}} N_{\text{H}}}$$

单个尘埃粒子和氢原子的质量分别为:

$$m_{\text{dust}} = \frac{4}{3} \pi a^3 \rho, \quad m_{\text{H}} = 1.008 \text{ u} = 1.674 \times 10^{-24} \text{ g}$$

又有:

$$\frac{A_{0.55 \mu\text{m}}}{N_{\text{H}}} \approx 3.52 \times 10^{-22} \text{ mag} \cdot \text{cm}^2$$

注意到 $A = \frac{2.5}{\ln 10} \tau = 1.086 \tau$, 且 $\tau = n \sigma l = C_{\text{ext}} N_{\text{dust}}$, 则有:

$$1.086 C_{\text{ext}} N_{\text{dust}} = 3.52 \times 10^{-22} \text{ mag} \cdot \text{cm}^2 N_{\text{H}}$$

即:

$$\frac{N_{\text{dust}}}{N_{\text{H}}} = \frac{3.52 \times 10^{-22} \text{ mag} \cdot \text{cm}^2}{1.086 \pi a^2 Q_{\text{ext}}}$$

于是有：

$$\begin{aligned}
 \frac{M_{\text{dust}}}{M_{\text{H}}} &= \frac{m_{\text{dust}} N_{\text{dust}}}{m_{\text{H}} N_{\text{H}}} \\
 &= \frac{(4\pi/3)a^3 \rho}{m_{\text{H}}} \cdot \frac{3.52 \times 10^{-22} \text{ mag} \cdot \text{cm}^2}{1.086\pi a^2 Q_{\text{ext}}} \\
 &= \frac{(4\pi/3)a^3 \rho}{1.674 \times 10^{-24} \text{ g}} \cdot \frac{3.52 \times 10^{-22} \text{ mag} \cdot \text{cm}^2}{1.086\pi a^2 Q_{\text{ext}}} \\
 &= 258.2 \text{ mag} \cdot \text{cm}^2/\text{g} \cdot \frac{a\rho}{Q_{\text{ext}}}
 \end{aligned}$$

21.3 (b) 代入 $a = 0.1 \mu\text{m} = 10^{-5} \text{ cm}$ 、 $Q_{\text{ext}} = 1.5$ 、 $\rho = 3 \text{ g/cm}^3$ 得：

$$\frac{M_{\text{dust}}}{M_{\text{H}}} = \frac{a\rho}{Q_{\text{ext}}} \cdot 258.2 \text{ mag} \cdot \text{cm}^2/\text{g} = 5.164 \times 10^{-2}$$